在 SnapDrive for UNIX 中支持多路径

Snapdrive for Unix

NetApp
October 26, 2021

This PDF was generated from https://docs.netapp.com/zh-cn/snapdrive-unix/aix/task_enabling_multipathing.html on October 26, 2021. Always check docs.netapp.com for the latest.

目□

在 SnapDrive for UNIX 中支持多路径□□ 1
□用多路径 1
刷新 DMP 路径的原因 4

在 SnapDrive for UNIX 中支持多路径□□

SnapDrive for UNIX 支持使用□准多路径□件解决方案□存□系□□行 FC 多路径□□。通□使用多路径，□可以在主机和存□系□之□配置多个网□路径。

平台	多路径解决方案
AIX	□于 FC • NativeMPIO • DMP

如果一个路径□生故障，□ FC 流量将□□在其余路径上□□。如果主机具有多个 LUN 路径，□需要多路径，并且它可以使底□路径□用□透明。如果配置并支持 SnapDrive for UNIX 操作配置文件中指定的多路径解决方案，□ SnapDrive for UNIX 将使用指定的多路径解决方案。

□用多路径

要在 SnapDrive for UNIX 中使用多路径，必□完成包括安装和□□ HBA 在内的□□。如果□使用的是非 NetApp 多路径解决方案，□必□从 HBA 供□商的网站下□□件和□用程序包。

□的系□必□□行当前版本的 FC 和 iSCSI 系□□件。

"NetApp 互操作性表工具"

□□

1. 在安装相□的 Host Utilities □件之前，□先安装支持的 HBA 。

   □用于 UNIX 的 SnapDrive 支持多供□商 HBA 。

2. □□ HBA 服□。

   如果 HBA 服□未□行，□在□行 SnapDrive for UNIX 命令□会□示以下□□消息，例如 SnapDrive storage create` 和 SnapDrive config prepare LUNs` ：

   ```
   0001-876 Admin error: HBA assistant not found
   ```

   □必□□□所需数量的路径是否已□□且正在□行。□可以使用 Host Utilities □件附□的 sanlun □用程序□□路径。例如，在 FC 多路径配置中，□可以使用 `sanlun fcp show adapter -v` 命令。

   在以下示例中，有□个 HBA 端口（ fcd0 和 fcd1 ）□接到□行正常的主机（端口状□）。□只能有一个 HBA 或 iSCSI □□程序，并且必□通□□目□ LUN 提供多个路径来配置多路径。

```
# sanlun fcp show adapter -v
adapter name: fcd0
WWPN: 50060b000038c428
WWNN: 50060b000038c429
driver name: fcd
model: A6826A
model description: Fibre Channel Mass Storage Adapter
(PCI/PCI-X)
serial number: Not Available
hardware version: 3
driver version: @(#) libfcd.a HP Fibre Channel
ISP 23xx & 24xx Driver B.11.23.04
/ux/core/isu/FCD/kern/src/common/wsio/fcd_init.c:Oct 18
2005,08:19:50
firmware version: 3.3.18
Number of ports: 1 of 2
port type: Fabric
port state: Operational
supported speed: 2 GBit/sec
negotiated speed: 2 GBit/sec
OS device name: /dev/fcd0
adapter name: fcd1
WWPN: 50060b000038c42a
WWNN: 50060b000038c42b
driver name: fcd
model: A6826A
model description: Fibre Channel Mass Storage Adapter
(PCI/PCI-X)
serial number: Not Available
hardware version: 3
driver version: @(#) libfcd.a HP Fibre Channel
ISP 23xx & 24xx Driver B.11.23.04
/ux/core/isu/FCD/kern/src/common/wsio/fcd_init.c:Oct 18
2005,08:19:50
firmware version: 3.3.18
Number of ports: 2 of 2 port type: Fabric
port state: Operational
supported speed: 2 GBit/sec
negotiated speed: 2 GBit/sec
OS device name: /dev/fcd1
```

如果在主机上□用了多路径，□可以□同一 LUN □示多个路径。□可以使用 `sanlun lun show all` 命令□□路径。在以下示例中，□可以□到指向同一 LUN 的多个路径（□： `/vol/vol1/lun` ）：

```
filer:          lun-pathname        device filename  adapter    protocol
lun size              lun state
    filer_1:        fish/vol/vol1/l1      hdisk36          fcs0
FCP       5m   (5242880)        GOOD
    filer_2:        fish/vol/vol0/lunKr   hdisk39          fcs0
FCP       100m (104857600)      GOOD
    filer_3:        fish/vol/vol0/testaix hdisk2           fcs0
FCP       200m (209715200)      GOOD
    filer_4:        fish/vol/vol1/l1      hdisk45          fcs1
FCP       5m   (5242880)        GOOD
```

3. 在 FC 配置中，使用交□机分区配置□主机 HBA 端口和目□端口□行分区。
4. 安装并□置相□的 FC 或 iSCSI 交□机。
5. □□ SnapDrive for UNIX 堆□要求。
6. 安装或升□□用于 UNIX 的 SnapDrive 。
7. □□ SnapDrive for UNIX 安装。
8. □到 `snapdrive.conf` 文件路径。
9. 在 `snapdrive.conf` 文件中配置以下配置□量：
   - ` 多路径□型 _`
   - `` default-transport_
   - ` fsttype_`
   - ` vmtype_`

     □于□台主机，多路径□型，□□□型，文件系□和卷管理器□型彼此相□。下表介□了所有可能的□合：

主机平台	默□□□□型	多路径□型	**FSType**	**vmtype**
AIX	FCP	□□	JFS2 或 jfs	LVM
nativempio	JFS2 或 jfs	LVM	DMP	VxFS

+ 上表提供了 ` multipathing-type_` ， ` default-transport_` ， ` fsttype_` 和 ` vmtype_` 配置□量支持的□。

10. 保存 `snapdrive.conf` 文件。

    SnapDrive for UNIX 会在□次□□此文件□自□□其□行□□。要使更改生效，必□重新□□ SnapDrive for UNIX 守□□程。

    - 相□信息 *

"NetApp 支持"

"NetApp 互操作性"

"《 AIX Host Utilities 6.0 安装和□置指南》"

刷新 DMP 路径的原因

在具有 FC 和 DMP 配置的主机上， SnapDrive storage delete - lun` 命令可能会挂起。

命令可能会因以下□件安装或配置不当而挂起：

- NTAPI
- Veritas 堆□（存□基□）
- 多路径□可□

□用，禁用或添加任何 FC 路径后，□必□正□刷新 DMP 路径信息。要刷新 DMP 路径，□按□序□行以下命令，如下所示。

* 平台 *	* 命令 *
AIX	• `cfgmg` • `vxdisk scandisk`

Copyright Information

Copyright © 2021 NetApp, Inc. All rights reserved. Printed in the U.S. No part of this document covered by copyright may be reproduced in any form or by any means-graphic, electronic, or mechanical, including photocopying, recording, taping, or storage in an electronic retrieval system- without prior written permission of the copyright owner.

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

THIS SOFTWARE IS PROVIDED BY NETAPP “AS IS” AND WITHOUT ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE HEREBY DISCLAIMED. IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice. NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp. The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S. patents, foreign patents, or pending applications.

RESTRICTED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (c)(1)(ii) of the Rights in Technical Data and Computer Software clause at DFARS 252.277-7103 (October 1988) and FAR 52-227-19 (June 1987).

Trademark Information

NETAPP, the NETAPP logo, and the marks listed at http://www.netapp.com/TM are trademarks of NetApp, Inc. Other company and product names may be trademarks of their respective owners.